

Power BI Assignment 3 — Parameters, Incremental Refresh, Semantic Model & RLS

Objective

Build a production-style Power BI semantic model using incremental refresh, parameters, field parameters, and row-level security. Use the InfoSec table to drive regional access.

Dataset

Use Task_3_Incremental_Refresh_Parameters_Semantic_RLS_Data.xlsx. FactTransactions contains 150,000 transactions across multiple years, DimStore contains store metadata, and InfoSec is a security mapping table containing user email, role and region.

Assignment Requirements

1. Create RangeStart and RangeEnd Power Query parameters with DateTime type.
2. Filter FactTransactions using RangeStart and RangeEnd in Power Query, preserving query folding where the source supports it.
3. Configure Incremental Refresh on FactTransactions. Use a practical archive period and refresh period, and document your choices.
4. Create a field parameter that allows the report user to switch a chart axis between Revenue, Profit, Quantity and Orders (or another sensible metric set).
5. Create a second parameter/parameter-driven interaction for a business scenario such as Top N, target threshold, or selected analysis dimension. Explain its purpose.
6. Build a clean semantic model with FactTransactions in the center and dimensions such as DimStore and any additional dimensions you create. Keep relationships simple and intentional.
7. Create core measures: Total Sales, Total Cost, Total Profit, Profit Margin %, Transactions, Units and Average Transaction Value.
8. Create dynamic RLS using the InfoSec table. The user should be filtered to their Region based on USERPRINCIPALNAME().
9. Handle the Executive row where Region = ALL. Executives should be able to see all regions; regional managers should only see their assigned region.
10. Create an InfoSec report tab showing the security mapping, current USERPRINCIPALNAME(), current role/region logic, and a small explanation of how RLS works. Do not expose sensitive real-world credentials.
11. Test RLS with at least three regional users and one Executive scenario using Power BI's role-testing features.
12. Publish the semantic model/report and document which settings must be configured in the Power BI Service for scheduled/incremental refresh and gateway/source connectivity, if applicable.

Required Deliverables

- PBIX/PBIP with semantic model and report.

- Screenshot or export showing RangeStart/RangeEnd and Incremental Refresh configuration.
- Working field parameter used in a visual.
- Working RLS role and InfoSec page.
- A short architecture/model diagram or screenshot.
- A one-page technical note explaining refresh strategy, parameters and security.

Evaluation Checklist

Area	Expected evidence
Incremental Refresh	Range parameters filter the fact query and policy is configured correctly.
Field Parameter	User can switch the analytical perspective of a visual without rebuilding the visual.
Semantic Model	Star-style relationships, correct cardinality and measures.
RLS	Regional users see only their region; Executive sees all regions.
InfoSec	Security mapping and current-user logic are clearly demonstrated.
Production readiness	Student explains refresh/service considerations and avoids hard-coded user rules.

Submission note: Keep the model clean, use meaningful names, avoid unnecessary calculated columns, and document important modeling decisions.